The relationship between grassland species richness and the management in long-established golf courses in Japan

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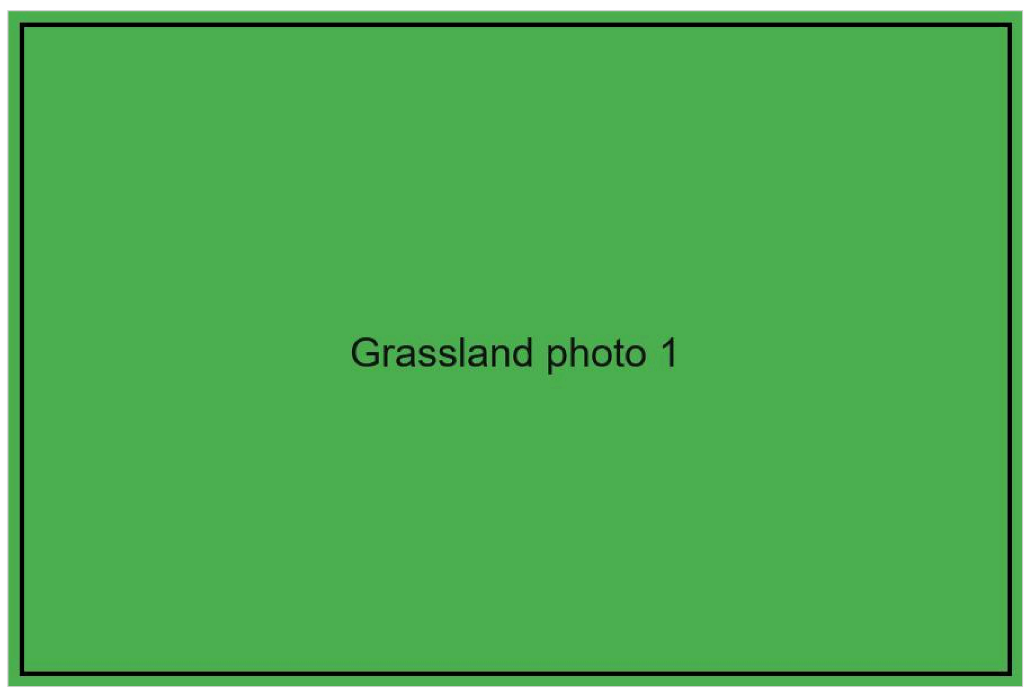
OBJECTIVES

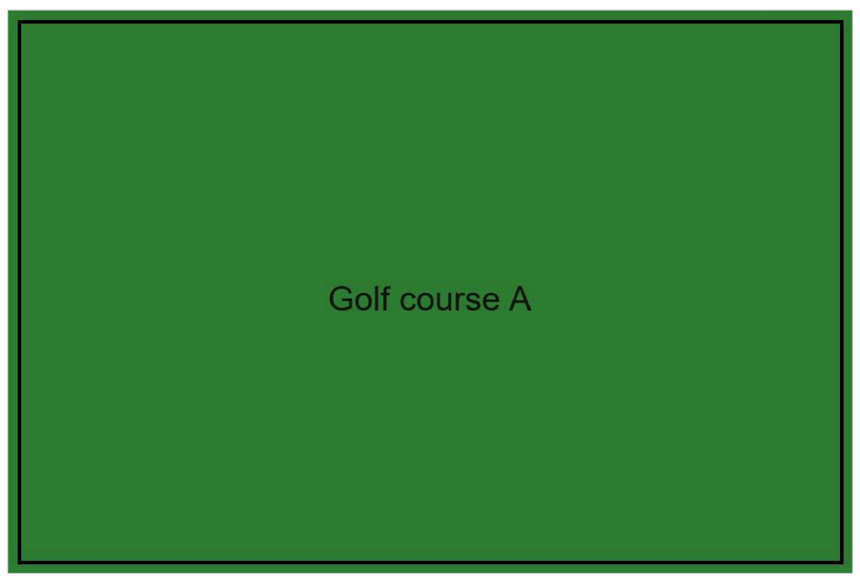
For conservation of grassland plants,to clarify the management methods and the availability oflong-established golf courses.

BACKGROUNDS

Semi-natural grassland is one of the most important habitats.Area and species diversity of semi-natural grasslands declinedby lifestyle change and abandonment.

Remaining semi-natural grasslands are confined toagricultural margins and commercial grasslands(long-established pastures, ski slopes, golf courses).





METHODS

Field investigations

- 4 golf courses (established in 1903-1956)
- 50 plots (1m x 1m) at rough or OB in each course
- Flora lists of grassland species in each golf course

Greenkeepers interview of management methods

- Frequency and height of mowing
- Application of fertilizer and Chemical herbicides

項目	内容
調査地	ゴルフ場4か所
方法	プロット調査（50 plots）

CONCLUSIONS

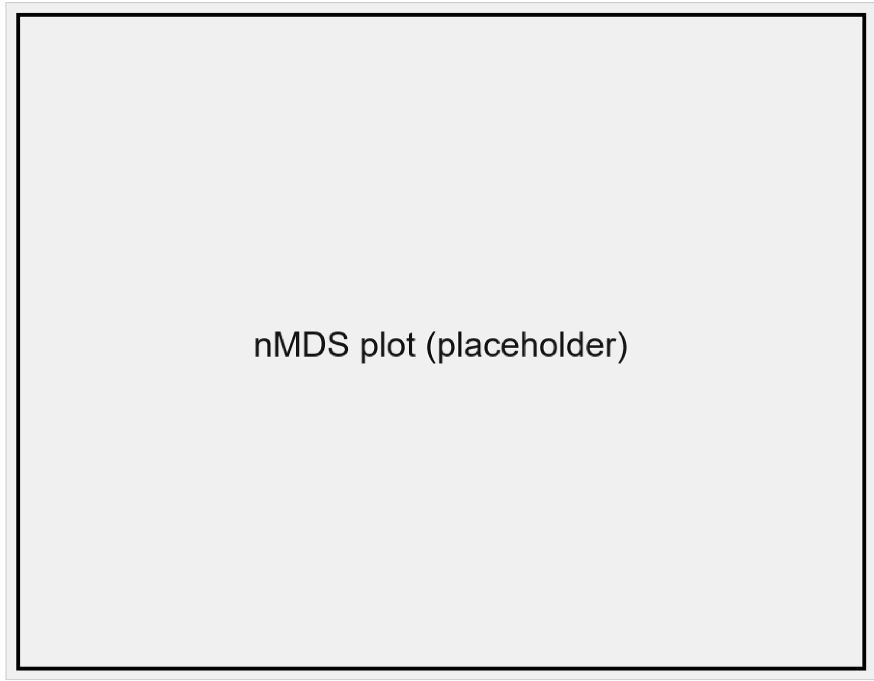
- Grassland species exist in long-established golf courses.
- Management of low frequency and height cutting height resulted in species rich grasslands.

★ Golf course in appropriate management can be seed source and refugia.

RESULTS

Species richness of each community

Community	Number of plots	Grassland Species (/m2)	All species (/m2)	Cutting height (cm)	Cutting Frequency (times/yr)
com1	57	6.1 ± 2.9	10.7 ± 4.7	12.9 ± 15.3	4.2 ± 6.1
com2	38	5.3 ± 2.7	10.6 ± 5.2	22.7 ± 19.2	4.4 ± 4.4
com3	37	3.9 ± 1.3	6.9 ± 3.1	5.1 ± 4.8	10.1 ± 5.3
com4	68	1.4 ± 0.7	2.4 ± 1.6	5.2 ± 5.0	14.6 ± 5.5



SUMMARY

Summary of each golf course

Golf course	Established year	Area (ha)	Total number of grassland species	Grassland species (Mean ± SD) (/m2)	All species (Mean ± SD) (/m2)
A	1903	20	79	6.5 ± 2.8	11.2 ± 4.6
B	1926	70	55	2.9 ± 2.0	5.2 ± 4.6
C	1930	55	78	4.4 ± 2.8	8.1 ± 5.1
D	1956	60	55	2.1 ± 1.5	4.0 ± 3.3

DOMINANT SPECIES